

Daniel Ayomide Olanrewaju

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Google DeepMind Scholar (AIMS, South Africa)

1. Research Interests

Quantum Computing (Hamiltonian simulation, ground state problems), Quantum Machine Learning(QML), Generative AI, Algebra

2. Education

African Institute for Mathematical Sciences, South Africa (2025–July 2026)

Master of Science in Mathematical Sciences with specialization in Artificial Intelligence for Science.

- Supervisors: Dr. Ryan Sweke and Dr. Kouhei Nakaji
- Thesis: *Generative Hamiltonian Simulation*

University of Ibadan, Nigeria (2017–2023)

Bachelor of Science in Mathematics

- Final Grade: First-Class Honours — CGPA: 3.66/4.0
 - Supervisor: Dr. Praise Adeyemo
 - Thesis: *Some Applications of Group Theory to Polynomial Equations*
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3. Publications

- Daniel, O. A. (2025). *Partial Symmetry Enforced Attention Decomposition (PSEAD): A group-theoretic framework for equivariant transformers in biological systems*. [arXiv:2507.14908](#)
 - Daniel, O. A. (2025). *A Comprehensive Computational Approach for Detecting Multilingual, Multicultural, and Multievent Online Polarization in English and Swahili*. DOI: [10.13140/RG.2.2.11612.35205](#)
 - Daniel, O. A. (2025). *Architecting a Scalable Recommender System: From Alternating Least Squares to Production-Ready API*. DOI: [10.13140/RG.2.2.11106.08649](#)
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4. Research Experience

Research Assistant, Lagos State University, Nigeria (2022)

- Assisted in data collection and analysis for research on partner support for exclusive breastfeeding in Ibadan North East LGA, Oyo State, Nigeria.

Research Assistant, University of Ibadan, Nigeria (2021–2022)

- Contributed to research on awareness and utilization of preconceptional folic acid and iron supplements among women in Lagos, handling data collection, analysis, and publication preparation.
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5. Work Experience

Machine Learning Specialist & Software Engineer, Lamzytech, Jos, Nigeria (2023–2026)

- Developed and deployed frontend, backend, and machine learning models for various applications, improving service delivery by at least 40% (see GitHub for details).
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6. Teaching Experience

Assistant Instructor, Data Science and Machine Learning, Lamzytech, Jos (2024)

Undergraduate Tutorial Teacher, University of Ibadan (2021–2023)

- Courses: Operations Research (MAT 353), Calculus and Trigonometry (MAT 121)

Group Leader, Peer Assisted Study Session, University of Ibadan (2019)

- Courses: Calculus and Trigonometry (MAT 121)

7. Service

Leadership and Committee Roles

- (2024) Financial Secretary, Financial Inclusion CDS, National Youth Service Corps (NYSC)
- (2021–2023) Member, Students Academic Committee, Department of Mathematics, University of Ibadan

Volunteering

- (2025) Volunteer – 1st AIMS Workshop and School on The Theory of Quantum Learning Algorithms. (AIMS South Africa)
- (2020, 2022) Student Intern – Lagos Bureau of Statistics; Volunteer Data Engineer for the Lagos State WASH Survey

8. Awards and Honors

- (2025) Google DeepMind Scholarship – AI for Science, AIMS South Africa
- (2024) Excellent Leadership Award – NYSC Jos-South LGA, Plateau State
- (2019, 2021, 2023) Dean’s Roll of Honor Award, Faculty of Science, University of Ibadan
- (2023) Award of Academic Excellence – Department of Mathematics, University of Ibadan

9. Talks and Seminars

- (2025) Participant – 1st AIMS Workshop and School on The Theory of Quantum Learning Algorithms. (AIMS South Africa)
- (2025) Participant – Online Senior Learning Seminar in Algebraic Geometry (Prof. Balázs Szendrői)

10. Certifications

- IBM – Introduction to Quantum Machine learning
- Google Quantum AI (Coursera) - Hands-on Quantum Error correction with Google Quantum AI
- University of Michigan (Coursera) – Python for Everybody Specialization
- DeepLearning.AI (Coursera) – Deep Learning Specialization, TensorFlow Developer Specialization
- University of California (Coursera) – Initiating and Planning Projects

11. Proficiencies

- Quantum computing: Qiskit, IBM Quantum hardware (Hamiltonian simulation, circuit compilation, noise-aware execution)
- Programming: Python, TypeScript, Java, MATLAB, R, CUDA C++

12. AI Projects

(Others on [GitHub](#))

- **Real-Time Fraud Detection System** — Designed and deployed a scalable machine learning pipeline for detecting financial fraud in real time. The system uses Gradio for the user interface, Dagster for workflow orchestration, and MLflow for model tracking and versioning. *Live Demo:* [Fraud Detection Dashboard](#)
- **Text Analytics and Linguistic Games Suite** — Implemented a text analysis and interactive Scrabble-style word game application using Python and Gradio. The system performs sentiment analysis, readability scoring, and token frequency statistics, while providing an educational language-based gaming experience. *Live Demo:* [Text Analyzer](#)

13. References

- **Google DeepMind Mentor:** Dr Kathryn Tunyasuvunakool | ktkool@deepmind.com
- **Thesis Supervisor:** Dr Ryan Sweke | rsweke@aims.ac.za